prolific writer, Alberuni compiled a list of his own works when he was sixty. He continued to write after that (he died in 1048 CE at the age of seventy-five), but his surviving list has 146 titles averaging ninety pages each. Nearly half of these works are on astronomy and mathematics, writes scholar George Saliba. Only twenty-two have survived, about half of which have been published.

Besides his work on India, Alberuni wrote another encyclopaedic work, *The Chronology of Ancient Nations*, containing an 'anthropological account of various cultures', both dead and living. It includes, for instance, the most detailed, 'scientifically reasoned' analysis of the Jewish calendar till then. As a set, writes Saliba, 'these two works preserve the best premodern description of the cultures Alberuni came to know'. In his other works, Alberuni 'developed new algebraic techniques for the solution of third-degree equations'. He defended 'the mathematical sciences against the attacks of religious scholars' by showing them how spherical trigonometry could accurately find the direction of Mecca.[9] Though Alberuni was an ace astronomer, he was often paid to be an astrologer. He seems to have played along reluctantly; in his own words, his faith in astrology was 'as weak as that of its least adherents'. His accomplishments include the following:

> In his works on astronomy, he discussed with approval the theory of the Earth's rotation on its axis and made accurate calculations of latitude and longitude. In those on physics, he explained natural springs by the laws of hydrostatics and determined with remarkable accuracy the specific weight of 18 precious stones and metals. In his works on geography, he advanced the daring view that the valley of the Indus had once been a sea basin.[10]

Alberuni published *India* after Mahmud died in 1030 and his son became king. With this change of despots, Alberuni's fortunes suddenly improved a lot. He now described Mahmud as 'having failed in the duties of a protector of art and science' at home.[11] He decried Mahmud's raids in *India*, especially in Mathura and Somnath, claiming that 'Mahmud utterly ruined the prosperity' of the regions he raided, created a hatred of the Muslims among the locals, and caused the Hindu sciences to retreat 'far away from those parts of the country conquered by us' to places 'where our hands cannot yet reach, to Kashmir, Benaras, and other places'. He was clearly not fond of his ex-boss.